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Effects of subsite alterations on substrate-binding mode in the active site of hen egg-white lysozyme

Izumi KUMAGAI, Katsumi MAENAKA, Futoshi SUNADA, Shigeki TAKEDA and Kin-ichiro MIURA
Department of Industrial Chemistry, Faculty of Engineering, The University of Tokyo, Bunkyo-ku, Japan

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The subsite structures in the active site of hen egg-white lysozyme were altered by site-directed mutagenesis. Replacement of Trp62, which is involved in apolar interaction with a sugar ring, and Asp101, which is hydrogen bonded to the same sugar ring in subsite B, led to a shift of the oligosaccharide-binding mode in the active-site cleft. Consequently, the double-mutant lysozyme (Trp62His, Asp101Gly) exhibited a drastic change of substrate-binding without any significant loss of enzymic activity. Conversion of Asn37, which is postulated to be involved in interaction with a sugar ring in subsite F, had a reverse effect on substrate binding. Nuclear magnetic resonance analysis of mutant lysozymes, in which Trp62 was replaced with Phe or His, suggested that these replacements not only altered the structure of the amino acid side chain at position 62 of the lysozyme, but also induced local structural changes around the residue at position 62.

Most enzymes which catalyze the hydrolysis of biopolymers such as polypeptides, polynucleotides and polysaccharides have a subsite structure in their active sites. A subsite is thought to be part of the substrate-binding site of an enzyme which interacts strongly or weakly with a residue of a polymer substrate.

Hen egg-white lysozyme catalyzes the hydrolysis of β-1,4 glycosidic bonds of polysaccharides constituting the peptidoglycan layer of bacterial cell walls, and polymers of *N*-acetyl-D-glucosamine and chitin. Crystallographic studies of this lysozyme have shown that the substrate-binding site of the enzyme contains six subsites, A–F, which are able to bind six saccharide residues, and that a trimer of GlcNAc binds to A, B and C subsites whereas a monomer of GlcNAc binds selectively to subsite C (Blake et al., 1967; Cheetham et al., 1992). Recently, a trisaccharide prepared from the peptidoglycan layer, MurNAc-GlcNAc-MurNAc (Mur, muramic acid), has been reported to bind to subsites B–D (Strynadka and James, 1991). Previous X-ray crystallographic study (Blake et al., 1967) proposed that Asp101 interacted with the acetamido group of a sugar residue in subsite A and with the primary alcohol group of a sugar residue in subsite B. However, a recent refinement of the lysozyme complex with $(GlcNAc)_3$ has revealed that the side chain of Asp101 is hydrogen bonded only to the hydroxyl oxygen atom at position 6 of a pyranose ring in subsite B (Cheetham et al., 1992). Three tryptophans, Trp62, Trp63 and Trp108, are located in the substrate-binding cleft and are believed to interact with sugar residues placed in subsites B–D. Of these, Trp62 is proposed to be involved in apolar interaction with a sugar ring bound to subsite B and hydrogen bonding to the hydroxyl oxygen atom at positon 6 of a sugar ring in subsite C (Blake et al., 1967; Cheetham et al., 1992). Trp62 is the residue most exposed to solvent and is very susceptible to chemical reagents (Hayashi et al., 1965). Selective chemical modifications of Trp62 produce marked changes in its enzymic activity. Conversion of Trp62 to other amino acid residues by site-directed mutagenesis leads to enhanced bacteriolytic activity (Kumagai et al., 1987; Kumagai and Miura, 1989).

From model-building studies on the interaction of sugar residues with subsites E and F, Asn37 is postulated to interact with a primary alcohol group of a sugar residue in subsite F (Blake et al., 1967). Most of the amino acid residues comprising the substrate-binding cleft are conserved among c-type lysozymes. However, the three residues, Asp101, Trp62 and Asn37, are substituted with other amino acid residues in some c-type lysozymes (Jollès and Jollès, 1984). On the basis of this molecular evolution of c-type lysozymes, we recently introduced these three replacements, either alone or in combination, into the active site of hen egg-white lysozyme by site-directed mutagenesis and found that mutant lysozymes showed the drastic alteration of bacteriolytic activity (Kumagai et al., 1992).

In the present study, aimed to elucidate the substrate-binding mechanism by which the enzymic activity was markedly affected following the conversion of Trp62, the hydrolytic activities of the mutant lysozymes, in which Trp62 was replaced with Tyr, Phe and His, toward a GlcNAc oligomer substrate were measured. We found not only that the hydrolytic activities of these mutant enzymes were substantial but that also significant differences existed in the hydrolytic patterns of the GlcNAc oligomer, especially with the His62 mutant lysozyme.

Furthermore, the effects of replacement of Asp101 or Asn37 with Gly on the hydrolytic activity toward a GlcNAc

Correspondence to I. Kumagai, Department of Industrial Chemistry, Faculty of Engineering, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo, Japan 113

Abbreviations. Mur, muramic acid; PNP, *p*-nitrophenyl; DQF-COSY, double-quantum-filtered two-dimensional correlation spectroscopy.

Enzymes. Hen egg-white lysozyme (EC 3.2.1.17); human lysozyme (EC 3.2.1.17).

oligomer were examined. These structural changes in subsites were also found to influence the oligosaccharide-binding mode. Finally, the double-mutated lysozyme, in which both Trp62 and Asp101 were replaced, exhibited a drastic alteration of the substrate-binding mode.

We also report the NMR analysis of two Trp62 mutant lysozymes.

EXPERIMENTAL PROCEDURES

Materials

Restriction endonucleases, DNA-modifying enzymes and DNA-sequencing kits were purchased from Takara Shuzo and Toyobo. Hen egg-white lysozyme and *p*-nitrophenyl penta-*N*-acetyl-*β*-D-chitopentaoside [PNP-$(GlcNAc)_5$] were purchased from Seikagaku Kogyo. The high-performance liquid-chromatography column (YMC-Pac A-014, 6 mm × 300 mm) was from Nishio Kogyo. S-Sepharose (fast flow) and Mono S HR 5/5 were obtained from Pharmacia. All other chemicals used were of reagent grade appropriate for biochemical use.

Construction of mutant lysozyme genes

Mutant hen lysozyme cDNA used in this study were prepared by site-directed mutagenesis as described previously (Kumagai and Miura, 1989; Kumagai et al., 1992). The cDNA carrying double mutations were constructed by utilizing the unique *Sac*I and *Hin*fI sites of the lysozyme cDNA. The mutations were confirmed by DNA sequencing analysis.

Production and purification of mutant hen egg-white lysozymes

Expression and purification of mutant hen lysozymes were carried out according to the procedures of Kumagai and Miura (1989), using the expression system in *Saccharomyces cerevisiae* AH22. The mutant lysozymes secreted into the yeast growth medium were purified by cation-exchange chromatography on S-Sepharose (fast flow), followed by a Mono S HR5/5 column. The size and homogeneity of the purified mutant enzymes were analyzed by sodium dodecyl sulfate/polyacrylamide gel electrophoresis (Laemmli, 1970). The enzyme concentration was determined spectrophotometrically ($A_{1\%}$ at 280 nm = 26.4 for the wild-type hen lysozyme), and the molar absorption coefficients of the mutant lysozymes containing Trp62Tyr, Trp62Phe and Trp62His were adjusted by factors of 0.89, 0.85 and 0.85, respectively (Kumagai and Miura, 1989).

Enzymic activities of the mutant hen egg-white lysozymes toward PNP-$(GlcNAc)_5$

PNP-$(GlcNAc)_5$ (0.26 mM) was incubated with 1 μg of each lysozyme at 37 °C in 50 μl 50 mM sodium acetate, pH 5.0. An aliquot (3 μl) was withdrawn at appropriate incubation intervals from the reaction mixture and analyzed by high-performance liquid chromatography, using a YMC-Pac A-014 (6 mm × 300 mm) column which was developed with acetonitrile/water (80: 20) eluent at a flow rate of 1 ml/min (Nanjo et al., 1988). PNP-$(GlcNAc)_n$ (n = 1−5), obtained by enzymic reaction, were detected at 300 nm. The concentration of each GlcNAc derivative was determined by calculating the area of each eluted peak. Under these conditions, linearity of the reaction time course was observed with a decrease of PNP-$(GlcNAc)_5$ and an increase of PNP-GlcNAc (PNP, *p*-nitrophenyl) and PNP-$(GlcNAc)_2$ up to 45 min.

For analysis of the pH dependence of the enzymic activity, 50 mM sodium acetate (pH 3.5−5.5) and 50 mM sodium phosphate (pH 6.0−8.0) were used. Other experimental conditions were the same as described above.

Structural analysis of the mutant hen egg-white lysozymes by NMR

^{1}H-NMR experiments were performed at 500 MHz on a Bruker AMX500 spectrometer at 37 °C. All samples were studied in H_2O containing 10% D_2O to provide the field/frequency lock signal. The wild-type lysozyme sample contained 4 mM lysozyme in a volume of 400 μl at pH 4.4. The Trp62Phe and Trp62His samples contained about 0.5 mM in a volume of 200 μl at pH 3.7. Proton chemical shifts were referenced with respect to the water signal, which is 4.6 ppm downfield from the proton resonance of 4,4-dimethyl-4-silapentan-1-sulfonate at 37 °C.

For the two-dimensional NMR experiments, 512 free-induction decays of 2 K data points were collected by using the time-proportional phase-incrementation method (Ernst et al., 1987) and spectra of 1 K × 2 K data points were obtained with zero-filling, after which a two-dimensional Fourier transformation with squared sine-bell window functions for both dimensions was performed. Double-quantum-filtered two-dimensional correlation spectroscopy (DQF-COSY) was performed according to Rance et al. (1983).

RESULTS

Replacement of amino acid residues in the active site of hen egg-white lysozyme

The mutant cDNA of hen egg-white lysozyme were prepared by site-directed mutagenesis and expressed in *S. cerevisiae* AH22 under the control of the glyceraldehyde-3-phosphate dehydrogenase promoter. The purified Trp62 mutant proteins were subjected to structural analyses by either amino acid sequencing or NMR spectroscopy. The replacement of Trp62 with Tyr was confirmed by tryptic peptide mapping of the Trp62Tyr mutant protein (Kumagai et al., 1992). These Trp62Tyr, Trp62Phe and Trp62His mutants were subjected to 500 MHz ^{1}H-NMR spectroscopic analysis. As shown in Fig. 8, disappearance of the N1H proton resonance of Trp62 was clearly observed in NMR spectra of the three mutant proteins (only spectra of Trp62Phe and Trp62His mutants are shown), indicating the site-specific replacement of Trp62.

Enzymic activities of the position 62 mutant hen lysozymes toward PNP-$(GlcNAc)_5$

Fig. 1a shows the hydrolytic patterns of a substrate, PNP-$(GlcNAc)_5$, with the wild-type lysozyme and the Trp62His mutant enzyme. The disappearance of the initial substrate and the increase of reaction products containing *p*-nitrophenyl groups at their reducing ends in the reaction mixture were monitored by estimating the amount of each GlcNAc oligomer by high-performance liquid chromatography. As shown in Fig. 1a, major reaction products detected by HPLC analysis of PNP-$(GlcNAc)_5$ with the wild-type lysozyme and all mutant lysozymes described in this paper were PNP-GlcNAc and PNP-$(GlcNAc)_2$. Linearity of the reaction time

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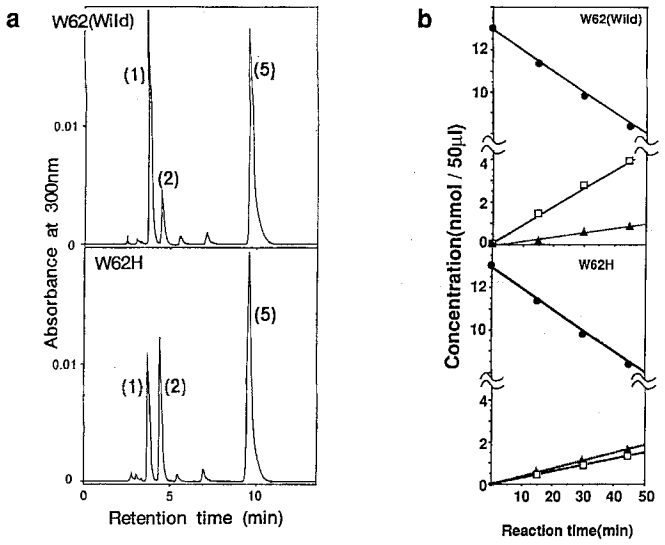

Fig. 1. Hydrolytic patterns of PNP-(GlcNAc)$_5$ catalyzed with the wild-type and Trp62His mutant lysozymes. (a) HPLC analysis of hydrolyzed products of PNP-(GlcNAc)$_5$ with the wild-type and mutant lysozymes. Hydrolyzed products were detected by HPLC on a YMC-Pac A-014 column (6 mm × 300 mm) developed with 80% CH_3CN at a flow rate of 1 ml/min; the absorbance unit on the scale was 0.02 at 300 nm. The reaction conditions were 50 mM CH_3COONa at pH 5.0, 0.26 mM PNP-(GlcNAc)$_5$, and 1 μg protein in 50 μl reaction mixture at 37°C. After 45 min incubation, 3 μl aliquots of the reaction mixture were subjected to HPLC analysis. Peak 1, PNP-GlcNAc; peak 2, PNP-(GlcNAc)$_2$; peak 5, PNP-(GlcNAc)$_5$. W62(wild-type Trp62), wild-type hen lysozyme; Trp62His, His-62 mutant enzyme. (b) Time-courses of hydrolytic reaction of PNP-(GlcNAc)$_5$ with the wild-type and mutant lysozymes. Experimental conditions were the same as described in the legend to Fig. 1a. At 0, 15, 30 and 45 min, the hydrolyzed products were analyzed and quantified by HPLC. (□) Increase of PNP-GlcNAc (1); (▲) increase of PNP-(GlcNAc)$_2$ (2); (●) decrease of PNP-(GlcNAc)$_5$ (5).

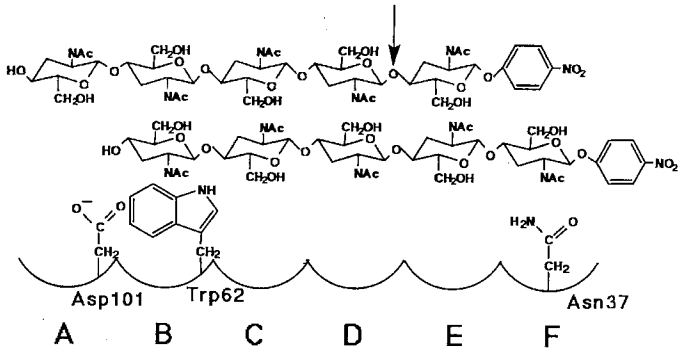

Fig. 2. Two productive binding modes of an oligosaccharide substrate in the active site creft of hen egg-white lysozyme. A–F show the subsites in the active site of hen lysozyme. An arrow indicates the cleavage site of the substrate. Asp101, Trp62 and Asn37, which were replaced with other amino acids in this study, are also represented.

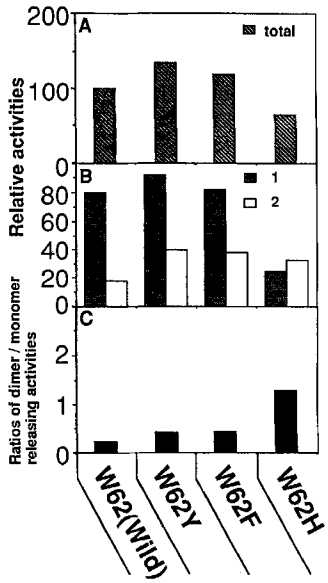

Fig. 3. Hydrolytic activities of PNP-(GlcNAc)$_5$ catalyzed with the wild-type and position-62 mutant lysozymes. The enzymic reactions were performed and analyzed as described in the legend to Fig 1. (A) Total activities, which were defined as the decrease of PNP-(GlcNAc)$_5$ with lysozymes, were expressed taking the total activity of the wild-type hen lysozyme as 100. (B) Increases of PNP-GlcNAc(1) and PNP-(GlcNAc)$_2$(2) were shown, also taking the total activity of the wild-type hen lysozyme as 100. (C) Ratios of PNP-(GlcNAc)$_2$-forming activity PNP-GlcNAc-forming activity are shown.

course was observed in both disappearance of the initial substrate and increase of these two products up to 45 min, under the conditions described in Experimental Procedures (Fig. 1b). The fact that the initial rate of disappearance of PNP-(GlcNAc)$_5$ was almost equal to the sum of the rates of increase of PNP-GlcNAc and PNP-(GlcNAc)$_2$ indicated that only cleavages of glycosidic bonds between sugar rings D and E, and between sugar rings C and D were detectable and that no transglycosylation occurred under the conditions we employed. Therefore, as shown in Fig. 2, in order to analyze the process of hydrolysis of PNP-(GlcNAc)$_5$ with the wild-type and mutant lysozymes, two productive binding modes were mainly considered.

Fig. 3 summarizes the initial rates of decrease of PNP-(GlcNAc)$_5$ and the increase of hydrolyzed products with the wild-type, Trp62Tyr, Trp62Phe and Trp62His mutant lysozymes. The Trp62Tyr and Trp62Phe mutant enzymes showed increased total activities, which were estimated from the disappearance of PNP-(GlcNAc)$_5$. However, the initial rates of increase of PNP-GlcNAc with these mutants were found to be almost the same as those for the wild-type enzyme. Therefore, these enhancements of the total hydrolysis rate were due mainly to the increased formation of PNP-(GlcNAc)$_2$. The ratios of PNP-GlcNAc and PNP-(GlcNAc)$_2$ formation were observed to be reduced by conversion of Trp62. Consequently, the ratio of PNP-GlcNAc and PNP-(GlcNAc)$_2$ formation with Trp62His mutant enzyme was reversed in comparison with that of the wild-type enzyme.

pH dependency of the hydrolytic activity of the position-62 mutant hen lysozymes toward PNP-(GlcNAc)$_5$

Fig. 4 shows the pH-activity profiles of PNP-(GlcNAc)$_5$ hydrolysis with the wild-type, Trp62Tyr, Trp62Phe and Trp62His mutant lysozymes. The optimal pH was observed to be around 5.0 for the wild-type, Trp62Tyr and Trp62Phe mutant enzymes. The optimal pH of the Trp62His mutant was closer to neutral pH. Also, the pH dependency of either

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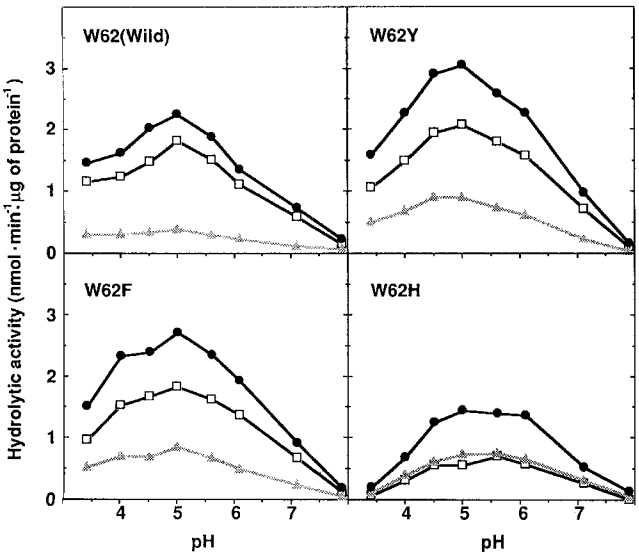

Fig. 4. pH dependency of the hydrolytic reaction of PNP-(GlcNAc)$_5$ with the wild-type and position-62 mutant lysozymes. The hydrolytic reactions were carried out almost as described in the legend to Fig. 1 expect for the buffer conditions. For pH 3.5−5.5, 50 mM sodium acetate and for pH 6.0−8.0, 50 mM sodium phosphate were used. (□) increase of PNP-GlcNAc(1); (▲) increase of PNP-(GlcNAc)$_2$; (●) decrease of PNP-(GlcNAc)$_5$(5).

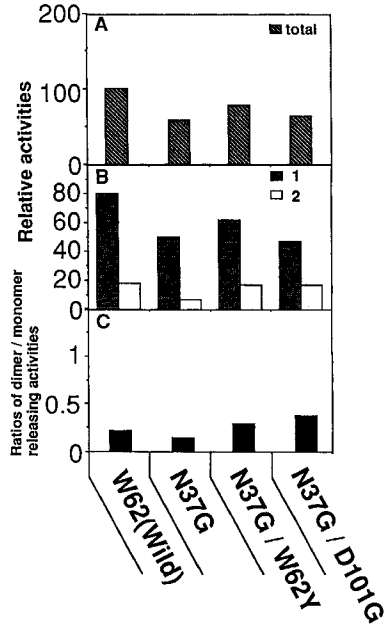

Fig. 6. Hydrolytic patterns of PNP-(GlcNAc)$_5$ catalyzed with the single-mutant and double-mutant lysozymes containing the replacement of Asn37. Experimental conditions and presentations were the same as described in the legend to Fig.3.

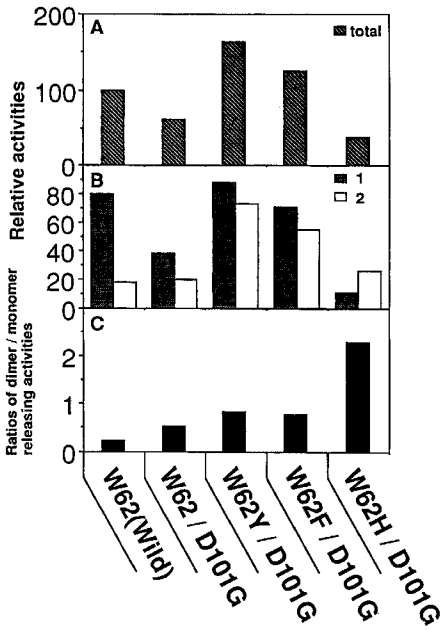

Fig. 5. Hydrolytic patterns of PNP-(GlcNAc)$_5$ catalyzed with the single-mutant and double-mutant lysozymes containing the replacement of Asp101. Experimental conditions and presentations were the same as described in the legend to Fig. 3.

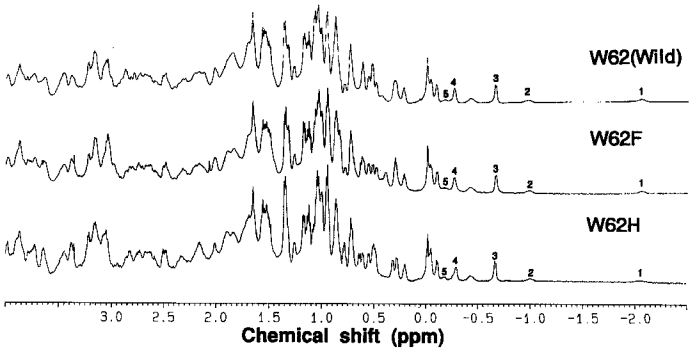

Fig. 7. High-magnetic-field region of 500-MHz spectra of the wild-type and two position-62 mutant lysozymes. Resonance labels 1 and 4 are CH(γ) and CH$_3$(γ) of Ile98 respectively, 2 is CH(β) of Met105, 3 is CH$_3$(δ) of Leu17, 5 is CH(β) of Lys116.

PNP-GlcNAc or PNP-(GlcNAc)$_2$ formation was found to be almost the same as that of total activity.

Effects of double mutations including those at position 62 on the hydrolysis of PNP-(GlcNAc)$_5$

As shown in Fig. 5, replacement of Asp101 with glycine was also found to influence the substrate-binding mode. Therefore, we examined the effects of mutation of Asp101Gly in addition to that at position 62 on the hydrolytic activity toward the chito-oligosaccharide substrate. The double mutant, Trp62Tyr Asp101Gly was found to show the highest hydrolytic activity toward PNP-(GlcNAc)$_5$ among the mutant lysozymes examined in this study, which also resulted from the increase of PNP-(GlcNAc)$_2$ formation. Furthermore, the substrate-binding mode of the Trp62His Asp101Gly mutant was markedly altered in comparison with that of the wild-type lysozyme.

Conversion of Asn37, which is proposed to be involved in F-site binding, showed a reverse effect to Asp101 on substrate binding. Replacement of Asn37 with Gly produced a relative increase of the substrate-binding mode which yielded PNP-GlcNAc (Fig. 6).

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Structural analysis of the position-62 mutant hen lysozymes by NMR measurement

The 500-MHz proton NMR spectra of the wild-type and two position-62 mutant lysozymes were recorded at 37°C and pH 4.4, conditions nearly optimal for the hydrolytic activity toward chito-oligosaccharide. These spectra of the Trp62Tyr and Trp62His mutant lysozymes were similar to that of the wild-type lysozyme reported by Redfield and Dobson (1988). The assignment of proton resonances, which was found to shift, was performed by sequential assignment using two-dimensional NMR analysis. The proton resonances of amino acid residues comprising the hydrophobic box (Ile98, Met105, Leu17, etc.) were scarcely shifted in these mutant enzymes (Fig. 7). In the low-field region of the spectra of Trp62 mutant lysozymes (Fig. 8), (a) the N1H proton resonance of Trp62 disappeared, which indicated the specific replacement of Trp62 with other amino acid residues, (b) the N1H proton resonances of Trp63 in the Trp62Phe and Trp62His mutant lysozymes were down-field and up-field shifted, respectively, and (c) the N1H proton resonances of Trp108 in the Trp62Phe and Trp62His mutant lysozymes were up-field and down-field shifted, respectively. Observations (b) and (c) suggested that replacement of Trp62 with either Phe or His had different influences on the local conformation of neighboring Trp63 and Trp108.

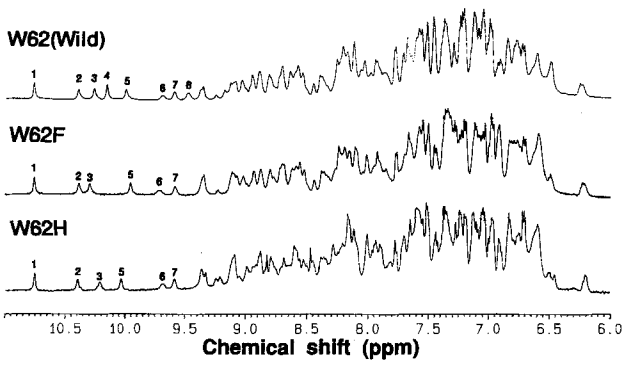

Fig. 8. Low-magnetic-field region of 500 MHz-NMR spectra of the wild-type and two position-62 mutant lysozymes. Resonance labels 1–5 are N1H of Trp123, Trp111, Trp63, Trp62 and Trp108, respectively, 6–8 are NH of Asp66, Gly 26 and Cys76, respectively. In these W62 mutant lysozymes, N1H of Trp62 have disappeared and NH of Cys76 are considerably shifted.

In order to obtain more detailed structural information about the Trp62 mutant lysozymes, they were subjected to two-dimensional NMR analysis using the DQF-COSY method. Fig. 9 summarized the chemical-shift changes of the backbone protons (NH and CαH) caused by conversion of Trp62 to His. By analysis of the DQF-COSY spectra of the Trp62His mutant lysozyme, the backbone proton resonances of more than 100 amino acid residues were assigned according to the assignment by Redfield and Dobson (1988). From these analyses, major changes in the chemical shift of backbone protons were observed in (a) a three-stranded antiparallel β sheet (residues 41–60), which contains one of the catalytic residues, Asp52, and (b) a loop region (residues 61–78) which contains Trp62. The largest changes were observed in a section of the loop region located at positions 72–76. As seen in Fig. 8, the NH resonance of Cys76 in the Trp62Phe and Trp62His mutant proteins were shifted to a considerable degree.

DISCUSSION

Using a chito-oligosaccharide, PNP-$(GlcNAc)_5$, as a substrate, the hydrolytic activities of the Trp62 mutant lysozymes, which had been found to exhibit markedly enhanced bacteriolytic activity (Kumagai and Miura, 1989), were characterized. Under our experimental conditions, two productive binding modes should be considered, one leading to monomer formation and the other to dimer formation. Replacement of Trp62 with either Tyr or Phe increased the total hydrolytic activity, which was due mainly to the increase of GlcNAc dimer formation. The binding constants of a substrate analogue, $(GlcNAc)_3$, to Trp62Tyr and Trp62Phe were observed to be several times lower than that of the wild-type enzyme (Kumagai et al., 1992, unpublished data). Therefore, the catalytic efficiency of these mutant lysozymes should be enhanced significantly. These replacements of Trp62 increased the population of a productive binding which corresponded to the mode of binding to subsites B–F.

Therefore, the replacement of Trp62 with Tyr, Phe and His might lead to reduced nonpolar interaction with a sugar ring in subsite B, and loss of hydrogen bonding to a sugar ring in subsite C, which in turn might change the specificity

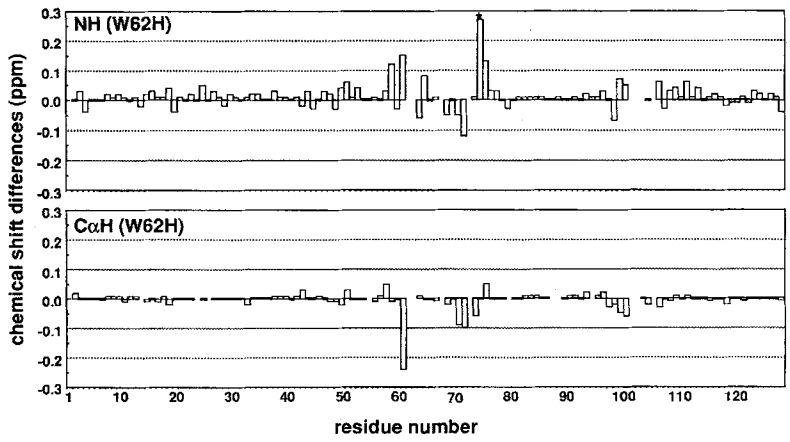

Fig. 9. The chemical-shift differences of the backbone protons by conversion of Trp62 to His. The open boxes indicate the degrees of the chemical-shift differences of NH and CαH. An asterisk indicates large chemical-shift difference of the NH of Leu75 (+0.57 ppm). The solid lines around 0 ppm shows no chemical-shift differences. No indication means unassigned protons in the present study.

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of each subsite for sugar residues slightly, whose conformations are tightly restricted in subsites B and C in the wild-type lysozyme. Such situations might make the mode of binding to subsites B−F preferential in the Trp62 mutant lysozymes. In fact, it is reported that human lysozyme, which has a Tyr residue at position 63 (numbered Tyr63 because of an insertion), bound chitotrioside in subsites A−C or in subsites B−D (Banjard, 1973; Shindler, M., 1977; Matsushima et al., 1990). The ratio of PNP-GlcNAc and PNP-$(GlcNAc)_2$ releasing activities from PNP-$(GlcNAc)_5$ with human lysozyme (Nanjo et al., 1988) seemed to be very similar to that with the Trp62Tyr mutant lysozyme. Furthermore, the replacement of Asp101 with Gly removed a hydrogen bond between the carboxyl group of Asp101 and the O(6) of a sugar ring in subsite B, which might lead to the reduction of specific interaction between the subsite and the sugar residue. Consequently, the double-mutant lysozyme, Trp62His, Asp101Gly, exhibited a marked preference toward the substrate-binding mode to subsites B−F. Although there is no experimental evidence for the involvement of Asn37 in the interaction with substrate oligosaccharide, this residue is postulated to interact with a sugar residue in subsite F (Blake et al., 1967). Recently, we replaced Asn37 with Gly and found that the mutation significantly affected the enzymic activity (Kumagai et al., 1992). The present observation that replacement of Asn37 with Gly increased the substrate-binding mode to subsites A−E might be taken as experimental evidence for the involvement of Asn37 in subsite F interaction.

In order to obtain the structural information on these mutant lysozymes, two Trp62 mutants (Trp62Phe and Trp62His) were subjected to analysis by NMR spectroscopy. The NMR data showed that the structure of the hydrophobic box was scarcely changed by replacement of Trp62, and the whole-molecule foldings of the position-62 mutant lysozymes were almost the same as that of the wild-type lysozyme. However, the local structures around position 62 of the Trp62Phe and Trp62His mutant lysozymes were found to be slightly different from that of the wild-type lysozyme. For example, the chemical shifts of N1H resonances of Trp63 and Trp108 in mutant lysozymes Trp62Phe and Trp62His were clearly different from that of the wild-type enzyme. In order to analyze the subtle structural changes of the position-62 mutant lysozymes, they were subjected to two-dimentional NMR analysis with DQF-COSY. The major changes in chemical shift were found to be localized in the three-stranded antiparallel β sheet (residues 41−60) and loop region (residues 61−78). The structural change in the antiparallel β sheet might cause a change in the chemical shift of the N1H proton of Trp108 in the position-62 mutant lysozyme, which hydrogen bonds to the carbonyl group of Leu56 located in the turn region of the antiparallel β sheet. The changes in chemical shift of the loop region (residues 61−78) also indicated that the replacement of Trp62 with Phe or His influenced the local folding (Miranker et al., 1991).

In particular, residues 72−76 of the loop region were markedly changed. It should be noted that this region was observed to change structurally when the wild-type lysozyme bound to $(GlcNAc)_3$ (Blake et al., 1981). Thus, it seems that replacement of Trp62 affects the local structure, which is important for substrate binding, and that not only the reduced nonpolar interaction with a sugar ring in subsite B but also such local structural changes lead to the alteration of both the substrate-binding constant and mode.

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